

When to Raise Taxes: State Dependence in the UK Tax Multiplier

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A tax rise costs about twice as much when the economy has slack. Using UK narrative tax measures from 1918 to 2018 and quarterly household consumption from 1955, I estimate the consumption response to tax changes that were not anticipated, and split it by whether unemployment was above or below its own recent average when the change took effect. Against a tax rise worth one per cent of annual GDP, consumption falls by 7.3 per cent at its trough when there is slack and by 3.2 per cent when there is not. The difference is estimated directly, with its own standard error, and is significant at eight of seventeen horizons. Consumption is the better outcome to use: it responds more than output, on three times as many horizons, and unlike the output response it holds when the single Budget on which the published UK multiplier rests is removed. The shock series reproduces Cloyne (2013) exactly before being extended, which fixes the conventions the estimate turns on. The aggregate record cannot separate one tax from another, cannot detect whether pre-announcement blunts the effect, and cannot say whose spending falls, and the paper measures each of those limits and names the household data that the distributional question needs.

Keywords: fiscal multipliers, narrative tax shocks, state dependence, local projections, household consumption

JEL classification: E32, E62, H20, H30

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When to Raise Taxes: State Dependence in the UK Tax Multiplier

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Abstract

A tax rise costs about twice as much when the economy has slack. Using UK narrative tax measures from 1918 to 2018 and quarterly household consumption from 1955, I estimate the consumption response to tax changes that were not anticipated, and split it by whether unemployment was above or below its own recent average when the change took effect. Against a tax rise worth one per cent of annual GDP, consumption falls by 7.3 per cent at its trough when there is slack and by 3.2 per cent when there is not. The difference is estimated directly, with its own standard error, and is significant at eight of seventeen horizons. Consumption is the better outcome to use: it responds more than output, on three times as many horizons, and unlike the output response it holds when the single Budget on which the published UK multiplier rests is removed. The shock series reproduces Cloyne (2013) exactly before being extended, which fixes the conventions the estimate turns on. The aggregate record cannot separate one tax from another, cannot detect whether pre-announcement blunts the effect, and cannot say whose spending falls, and the paper measures each of those limits and names the household data that the distributional question needs.

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1 Introduction

Governments choose when to raise taxes. The advice they receive rarely turns on that choice, because the empirical literature mostly reports a single number: the average response of output to a tax change, estimated over whatever sample is available. If the response is the same whenever the change lands, the timing is a political question and not an economic one.

This paper asks whether it is the same, and finds that it is not.

The exercise is possible because of a long narrative record. Romer and Romer (2010) identify legislated US tax changes from the contemporaneous documents and code the motivation behind

*Working paper. I thank James Cloyne for making the underlying narrative dataset public, and Cloyne, Dimsdale, Postel-Vinay and Hürtgen for depositing the interwar measures and the macroeconomic panel used here, without which this paper could not exist. A companion paper, *The anticipation content of fiscal policy*, builds the measure-level record this paper estimates on.

each, so that changes made for reasons unrelated to the state of the economy can be separated from those made in response to it. Cloyne (2013) builds the corresponding UK account for 1945 to 2009, and it has since been extended back into the interwar period by Cloyne et al. (2023) and forward to 2020 by Hürtgen et al. (2024), with the combined series used at century length by Cloyne et al. (2025). A companion paper (Conway, 2026) extends the measure-level record from Budget 2004 to Autumn 2018 and shows that the interval between announcing a UK tax change and its taking effect has roughly quadrupled since 1945, so that the share of the tax impulse arriving with more than four months of notice has risen from 0.18 to 0.70.

That finding shapes what can be done here. A measure announced two years before it takes effect is not a surprise on the day the money moves, and treating it as one estimates something other than a causal response. The projections below are therefore run on measures whose announcement and effect nearly coincide, which is a minority of the modern record and the price of the estimate meaning anything.

The question of whether the multiplier depends on the state of the economy is not new. Auerbach and Gorodnichenko (2012) find larger output responses to fiscal policy in US recessions, and Ramey and Zubairy (2018) dispute the finding for government spending using a longer sample and a different treatment of the news. Neither settles the question for tax changes, and neither is about the United Kingdom. What this paper adds is a UK answer, on household consumption rather than output, from a narrative series that reproduces the published one exactly before extending it.

The result is that against a tax rise worth one per cent of annual GDP, household consumption falls by 7.3 per cent at its trough when unemployment is above its own recent average, and by 3.2 per cent when it is not. Expressed per pound raised, the same tax rise costs roughly £4.50 of household spending in a slack economy and roughly £2 in a tight one. The gap is estimated directly rather than inferred from two separate regressions, and is significant at eight of seventeen horizons.

Section 5 reports what the aggregate record cannot support. It cannot say which tax matters, because most instruments are one or two episodes. It cannot say whether pre-announcement blunts a tax change, because the test lacks the power to detect an effect even as large as the one it is being compared against. And it cannot say who bears the cost, because splitting a measure by the incidence of its instrument produces two series that move together almost perfectly, so the distributional answer needs households observed directly, and the section names the survey that can carry it.

2 Data

2.1 The narrative record

Three public sources, all carrying announcement dates, implementation dates, a revenue costing and the same motive classification, are chained into one measure-level record.

The exogeneity restriction is the identifying assumption, and it comes from the original coders: measures are used only where the coder judged the motivation unrelated to the current state of the economy. Unlike the companion paper, which studies how long the policy process

Table 1: The measure-level record

Period	Source	Datable measures	Of which exogenous
1918–1938	Cloyne et al. (2023), Zenodo, CC-BY	204	89
1945–2009	Cloyne (2013)	1,221	1,221
2004–2018	Conway (2026)	173	173

Exogenous counts are of measures usable in estimation. 1939 to 1944 is absent from every source and 1921 contains no datable measure, so a century-long specification carries holes that estimation must handle rather than patch. The interwar block carries no tax-type field and so cannot enter any instrument-level specification.

takes and therefore needs no such restriction, a causal estimate does need it.

One design choice differs from the companion paper. That paper restricted attention to household-relevant measures, which required Cloyne’s `Group` field, which his own documentation disclaims. The interwar block has no such field at all, so the restriction cannot be carried back, and dropping it also matches the construction of the published exogenous series. The household-restricted variant is built alongside and the choice does not change what follows.

2.2 Reproducing the published series before extending it

A narrative shock series is a set of conventions as much as a set of measures, and small differences in those conventions matter more than they look. Before extending anything, the construction here was made to reproduce Cloyne (2013) exactly.

His aggregation code, published with his dataset, pushes an implementation date forward by 45 days before taking its calendar quarter, replaces an implementation date that precedes its own announcement with the announcement date, treats a measure as a surprise when it takes effect within 90 days, and divides by annualised nominal GDP. Rebuilding his 1955 to 2009 surprise series from his raw file under those rules returns it at a correlation of 1.0000, with a maximum difference of zero across all 220 quarters.

That exercise then prices each convention. Table 2 switches one at a time away from his baseline and reports the correlation with his published series.

Table 2: What each dating convention is worth

Construction	Correlation with the published series
His conventions exactly	1.0000
A 120-day surprise threshold instead of 90	0.9236
A half-month quarter rule instead of his 45-day shift	0.9316
Retroactive dates kept rather than replaced	0.9648
A different nominal GDP vintage	0.9990
All four differences together	0.8643

The GDP vintage is irrelevant and the threshold matters most. The companion paper uses the 120-day rule for reasons specific to the British fiscal calendar and keeps it; this paper uses the 90-day convention, because comparability with the published multiplier literature is what matters here.

2.3 Outcomes

The macroeconomic panel is the one Cloyne et al. (2025) estimate on, taken from their replication deposit: quarterly real and nominal GDP, the deflator, unemployment, Bank Rate, the exchange rate, prices, revenue, expenditure and the deficit, 1918 to 2020, sourced from Thomas and Dimsdale (2017). Using their outcomes rather than assembling my own makes every comparison with the published estimates like for like. Their narrative government spending shocks, over the same span, are carried as a control.

Household consumption is not in that deposit. It is taken from Thomas and Dimsdale (2017) directly, which is the same underlying source, so consumption and output come from one vintage rather than two. Their real GDP and the panel’s agree at a correlation of 1.0000 on levels and 0.9999 on growth. The expenditure components begin in 1955Q1 and version 3.1 ends in 2016Q4, so consumption work runs over 248 quarters against 1945 to 2018 for output.

2.4 Identifying variation

Table 3 reports the identifying variation, which is scarce in the modern period.

Table 3: Standard deviation of the quarterly shock, per cent of annual GDP

Era	Quarters	Unanticipated	Non-zero quarters	News
1920–38	76	0.227	14	0.012
1945–79	140	0.259	41	0.317
1980–99	80	0.158	40	0.125
2000–18	76	0.046	35	0.142

The surprise series keeps roughly the same number of events throughout and loses four-fifths of its amplitude, so the surprises kept coming and stopped being large. That is the companion paper’s central finding seen from the estimation side, and it is why no result below is reported for the modern period alone.

3 Consumption responds more than output

For horizon h , the projection is

$$100 \times (\log y_{t+h} - \log y_{t-1}) = \alpha_h + \beta_h s_t + \Gamma_h(L) + \varepsilon_{t+h},$$

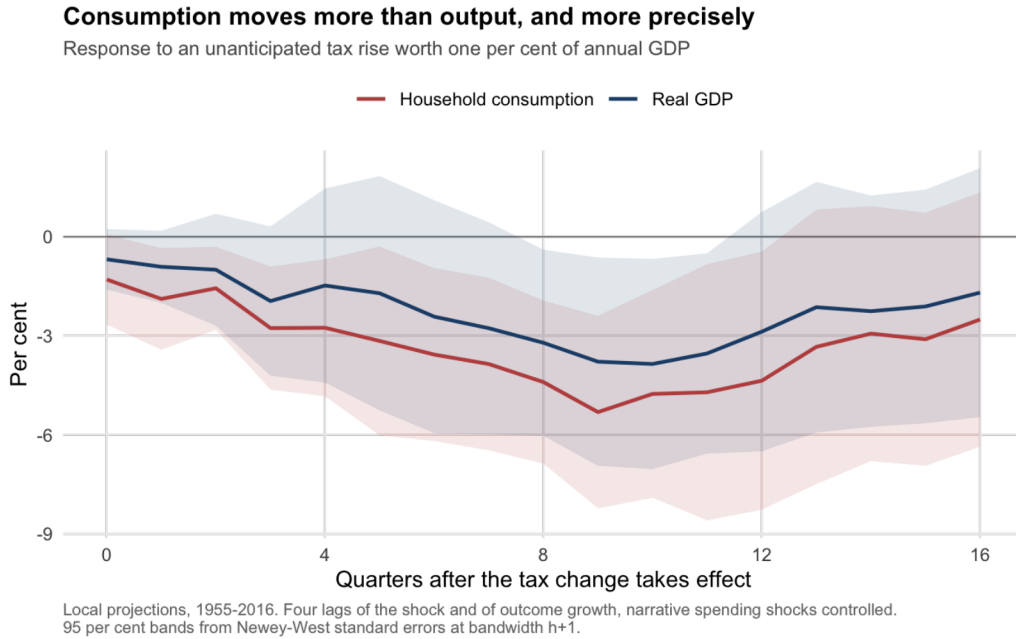
following Jordà (2005), where s_t is the unanticipated tax shock as a per cent of annual GDP, and the controls are four lags of the shock, four lags of outcome growth, and the contemporaneous narrative spending shock. Standard errors are Newey-West at bandwidth $h + 1$. A positive shock is a tax rise, so a contractionary response is $\beta_h < 0$, and β_h is the multiplier in the sense of Romer and Romer (2010).

Run on output, the estimate is respectable and fragile. Run on consumption, it is larger and holds.

Consumption is 61 per cent of GDP and moves more than output in absolute terms, so households carry more than their share of the adjustment. Figure 1 shows both paths.

Table 4: Response to an unanticipated tax rise worth one per cent of annual GDP

Outcome	Trough	Quarter	95 per cent interval	t	Significant horizons
Real GDP	-3.86	10	$[-7.04, -0.67]$	-2.37	4 of 17
Household consumption	-5.31	9	$[-8.23, -2.40]$	-3.57	12 of 17

**Figure 1:** Consumption responds more than output, and on more horizons.

The stronger reason to prefer consumption is robustness. The published UK tax multiplier rests heavily on one Budget. In Cloyne’s own 1955 to 2009 surprise series the June 1979 Budget, which raised VAT to 15 per cent, is the largest single observation at +1.596 per cent of GDP, and removing that quarter takes his aggregate output response from $t = -2.45$ with four significant horizons to $t = -1.46$ with none, and the same is true of the series built here and of the 2025 revision. The consumption response holds without it: removing the same quarter leaves a trough of -3.78 with $t = -2.27$, so the sign, most of the magnitude and the significance all stand.

That single Budget deserves a further remark. It enters at +1.596 in the 2013 coding and at +0.377 in the 2025 revision, so the largest postwar observation in this literature moved by a factor of four between two vintages of the same authors’ data. Any estimate that leans on it is leaning on a coding judgement, which is a reason to prefer estimates that do not.

4 The same tax rise costs twice as much in a slack economy

The state variable is an indicator for unemployment above its own seven-year backward moving average. It is backward-looking by construction, so it uses only information a policymaker had at the time, and it requires no output-gap estimate or filter. It puts 149 of the 248 usable quarters in slack.

Two separate estimates that differ do not by themselves establish the difference, so Table 6

Table 5: Consumption response by the state of the economy

State	Trough	Quarter	95 per cent interval	t	Significant horizons
Economy has slack	-7.31	11	$[-9.55, -5.06]$	-6.38	14 of 17
Economy does not	-3.16	8	$[-6.53, +0.20]$	-1.84	4 of 17

The narrative spending control is identically zero within these subsamples and is dropped from them. Removing the largest single quarter in each state leaves the slack trough at -7.37 with $t = -4.32$.

estimates the difference itself, by interacting the shock with the state indicator, so that the gap carries its own standard error.

Table 6: The difference between the two states, estimated directly

Quarters after the change	Slack minus tight	Standard error	t
4	-0.41	1.78	-0.23
8	-3.50	1.95	-1.80
9	-5.44	1.69	-3.22
11	-8.25	2.14	-3.86
12	-8.16	2.30	-3.54
16	-6.47	2.46	-2.63

Significant at eight of seventeen horizons. The gap opens from about the fifth quarter and is widest at eleven, which is also where the slack response troughs.

Figure 2 shows what the two states look like. The paths are indistinguishable for the first four quarters and separate after that: the slack response keeps falling and stays down, while the tight response troughs shallowly at two years and returns to zero.

Expressed the way a Chancellor would want it, and taking consumption at 61 per cent of GDP, a tax rise of one per cent of GDP is followed at its worst by a fall in household spending of about 4.5 per cent of GDP when there is slack, against about 2 per cent when there is not. The trough arrives at two to three years in both cases, so this is a claim about the medium run.

4.1 What is not shown

The multiplier does appear to fall as the pre-announcement regime takes hold, from a trough of -4.52 over 1955 to 1985 to -2.97 over 1986 to 2016. The late interval spans zero and the two overlap heavily, so this is a direction rather than an estimate.

A second bridge to the companion paper carries less weight than it appears to. If tax changes are now scheduled years ahead, a Chancellor might be unable to see the state a measure will land in, which combined with the result above would be a first-order cost of pre-announcement. The correlation between slack at announcement and slack at implementation does fall with notice, from 0.995 for measures taking effect within 90 days to 0.806 for those taking more than a year, and the typical state moves by 0.74 points rather than 0.03. That is real and it is modest: unemployment is persistent enough that even two years out a Chancellor mostly knows the state.

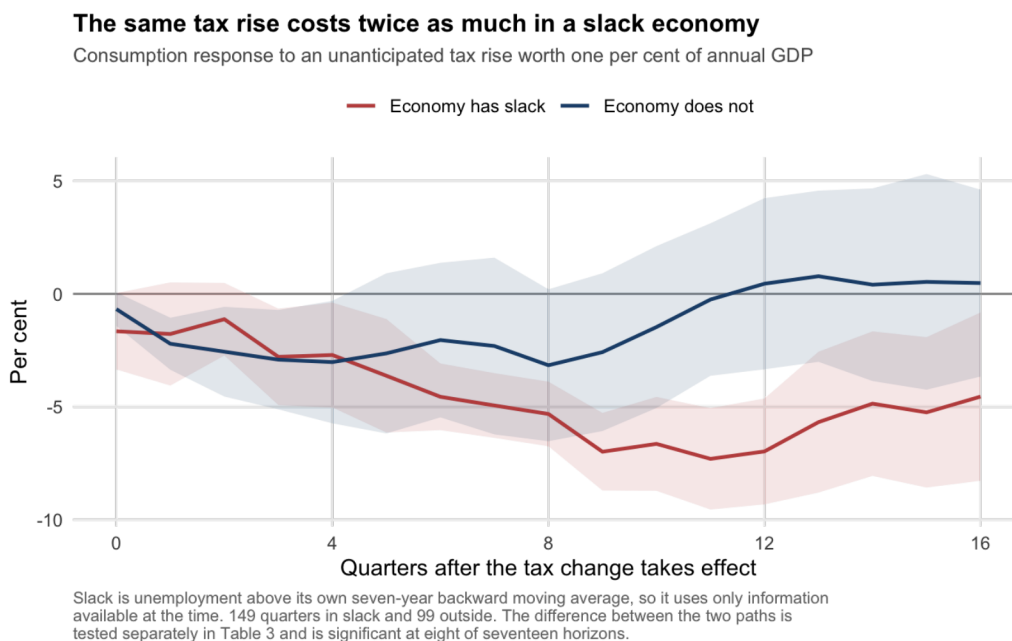


Figure 2: Consumption response by state, with 95 per cent bands.

5 What the record cannot support

Three further questions lie beyond the aggregate record, and the number that decides each is given below.

Which tax. Instrument-level multipliers look estimable on measure counts, with 449 income tax measures, 350 excise and 175 corporate. They are not estimable on variation. Eighty-four per cent of the absolute variation in the VAT surprise series sits in three quarters and the largest is June 1979; capital gains is 64 per cent concentrated and corporation tax 47 per cent. The resulting estimates are single episodes extrapolated to a shock worth one per cent of GDP, which for capital gains is a fifty standard deviation extrapolation, and they duly return implausible magnitudes. The UK narrative record cannot support a tax-by-tax decomposition.

Whether notice blunts the effect. The natural sequel to the companion paper is that pre-announced tax changes move consumption less, because households have already adjusted. The announcement-dated series returns no detectable response at any horizon. That null is uninformative: the minimum detectable effect of the design, the smallest true response it would find four times in five, is 10.5 per cent against a surprise response of 5.3. The test cannot distinguish no announcement effect from an announcement effect exactly as large as the surprise.

Who bears the cost. If a pound taken from a household with no buffer cuts spending by more than a pound taken from a household with savings, the aggregate response above conceals a distribution, and two tax packages of the same size can have different costs depending on where they land, which is the mechanism Cloyne et al. (2025) and the heterogeneous-agent literature would predict. The ONS Effects of Taxes and Benefits tables report, for every year from 1977, what each income decile pays in income tax, National Insurance, council tax, VAT

and each duty, and 1,036 measures carrying 83 per cent of gross value can be placed on that cash incidence; corporation tax, capital gains, inheritance and North Sea measures cannot, because the tables never allocate them to households. The gradient is real and correctly ordered: the bottom five deciles bear 0.36 of duties, 0.31 of VAT, 0.19 of stamp duty, 0.16 of National Insurance and 0.15 of income tax, so duties lean more than twice as far down the distribution as income tax does, as Figure 3 shows. Splitting each measure into a bottom-borne and a top-borne part and projecting consumption on each gives troughs of -14.3 and -5.1 , the right sign and the expected ordering, and it is not an estimate of heterogeneity: the two series correlate at 0.93 because they are slices of the same measures, the bottom-borne interval runs from -31.3 to $+2.7$, and standardised to a one standard deviation shock the two responses are 0.60 and 0.65 per cent of consumption. Slicing one shock cannot identify heterogeneity, because nothing in the data separates the slices, and the identification has to come from households observed directly. The Living Costs and Food Survey and its predecessor the Family Expenditure Survey record household expenditure from 1961 and allow the consumption response to be estimated by position in the income or liquidity distribution; the specification is written, the aggregate it decomposes is estimated above, and the household estimation is the next step of this work.

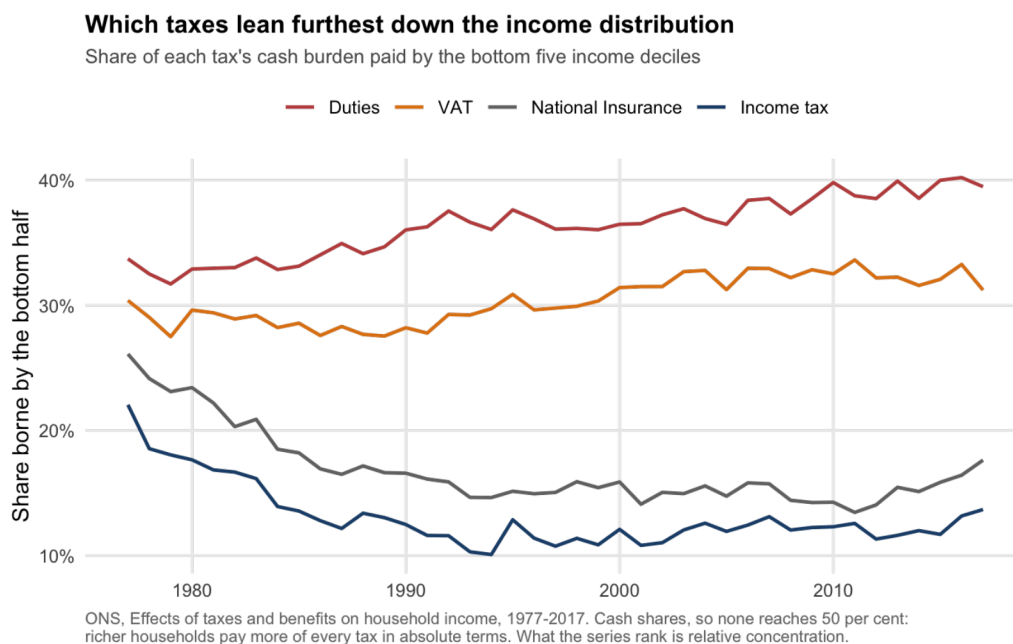


Figure 3: Share of each tax's cash burden borne by the bottom five income deciles.

6 Conclusion

The timing of a tax rise is an economic decision as well as a political one. On UK data from 1955 to 2016, the same unanticipated tax rise is followed by roughly twice the fall in household consumption when unemployment is above its recent average as when it is below, and the difference is large enough, and estimated precisely enough, to matter for the choice of when to consolidate.

Two qualifications belong with that. The result is about the medium run, with the gap

opening after a year and widest at eleven quarters, so it says nothing about the quarter of impact. And it is a statement about the average tax rise, because the record cannot separate one instrument from another and cannot yet say whose spending falls.

A second point may outlast the estimate. The published UK tax multiplier depends on a single Budget in June 1979 whose coded value moved by a factor of four between two vintages of the same dataset, and the output response disappears when that Budget is removed while the consumption response holds. Anyone estimating fiscal multipliers on this record should check which of their results are statements about the British economy and which are statements about that one Budget.

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